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AI PRINCIPLES AND PRACTICE · WEEK 7

Week 7 Summary

How a machine learns, and how it is wired: everything to keep, in plain English

IN PLAIN ENGLISH

How to use this summary. The ten points below follow the lecture in order. Read them (about 20 minutes), try the ten questions without looking (about 15 minutes), then mark them. Re-read only the points you got wrong. Everything here is something the lab, the final project and the viva build on.

KEY IDEA

If you remember one thing. A machine learns by guessing, checking, and nudging its numbers, thousands of times. Backpropagation is how it works out which numbers to nudge: one walk backwards from the mistake, multiplying as it goes.

Part 1 The ten points

1.1 A model is a formula with dials

A model is a formula whose numbers can be turned, like price = a × size + base price ($\hat{y} = ax + b$). The numbers it learns are its **parameters**, or dials; the settings you choose before training, such as the step size or the number of layers, are **hyperparameters**. **Learning** means turning the dials until the guesses match the real answers. With the three houses (60 m² for €180k, 90 m² for €240k, 120 m² for €300k) and a base price of €60k, the dial a = 2000 fits all three exactly.

1.2 The loop: guess, check, blame, nudge

Every model in the module learns with the same four steps. **Guess**: make a prediction with the dials you have (the forward pass). **Check**: boil all the mistakes down to one number, the loss. **Blame**: work out how much each dial caused the mistake (the backward pass). **Nudge**: move every dial a little, so the next guess is less wrong (the update). Then repeat, thousands of times.

1.3 One neuron: weigh, add, bend

A neuron multiplies each input by a weight, adds them up with a bias, and bends the total: $z = w_1x_1 + w_2x_2 + w_3x_3 + b$, then $\hat{y} = \sigma(z)$. In the umbrella example the score is $0.5 \times 1 + 2 \times 0.8 + (-1) \times 0.2 - 1 = 0.9$, and the S-curve (sigmoid) turns 0.9 into 71% sure. A negative weight means a clue argues against. One neuron with the S-curve is logistic regression from Week 2, and it can only ever draw **one straight line**: on the two moons it gets about 86% right.

1.4 ReLU, and why every neuron needs a bend

$\text{ReLU}(z) = \max(0, z)$: the output is the score if it is positive, otherwise 0. It works like a one-way valve. The bend matters because straight-line layers stacked without one collapse into a single straight line, $W_2(W_1x) = (W_2W_1)x$, so depth buys nothing. With a bend, a neuron can switch off for some inputs and on for others, and that is what lets a network make shapes.

1.5 A hidden layer moves the dots

Four dots (XOR: “one switch on, but not both”) cannot be separated by any straight line, so one neuron can never learn them. Two hidden neurons can: $h_1 = \text{ReLU}(x_1 + x_2)$, $h_2 = \text{ReLU}(x_1 + x_2 - 1)$, answer $= h_1 - 2h_2$ gives 0, 1, 1, 0. In the new numbers (h_1, h_2) both orange dots land on the same spot, (1, 0), and one straight line works. **A hidden layer moves the dots so that a straight line works.**

1.6 The forward pass and the dials

Asking a network something is the **forward pass**: numbers flow left to right, layer by layer, each layer doing $h = \text{ReLU}(Wx + b)$. The lecture’s network, $2 \to 16 \to 16 \to 1$, has 48 + 272 + 17 = **337 dials**. As one formula, $\hat{y} = \sigma(W_3\ \text{ReLU}(W_2\ \text{ReLU}(W_1x + b_1) + b_2) + b_3)$, read from the inside out.

1.7 The loss: one number for how wrong

For number answers, the loss is the **average squared miss**, $L = (1/n)\ \Sigma\ (\hat{y}_i - y_i)^2$: squaring stops misses cancelling and makes big misses count much more. With a = 1000 the house misses of 60, 90 and 120 give (3,600 + 8,100 + 14,400) ÷ 3 = 8,700. For yes/no answers, the loss is **cross-entropy**: minus the log of how sure the model was of the right answer. 90% sure and right costs 0.11; 50/50 costs 0.69; 5% on the right answer costs 3.00. Judge every model on data it has never seen: a big gap between practice and test scores means it **memorised**.

1.8 Gradient descent and the step size

New dial = old dial − step size × slope ($\theta \leftarrow \theta - \eta\ \partial L/\partial\theta$). The slope (the gradient) says which way is uphill and how steeply; the minus sign means “go downhill”. From a = 1000 with step size 40:

1000 − 40 × (−17.4) = 1696, and five steps reach 1997, nearly the valley floor at 2000. The steps shrink on their own as the ground flattens. The step size (the learning rate) is the first thing to check: too small crawls, too big bounces, far too big explodes to `nan` without any error message.

1.9 Blame and backpropagation

A dial's **blame** is how much the loss moves when you nudge it: with w = 3, x = 2, b = 1 and truth 5, nudging w to 3.001 moves the loss from 4.000 to 4.008, so the blame is 8. Effects along a chain multiply like gears (the **chain rule**): w moves the guess 2×, the guess moves the loss 4×, and 2 × 4 = 8. **Backpropagation** finds every dial's blame in one walk backwards, with three rules: blame at the end = guess − truth; a weight's blame = what came in × blame going out; blame is handed back through the same weights, and none passes through a switched-off neuron. Nudging is too slow to train with, but it checks backprop perfectly (a **gradient check**).

1.10 The loop, running

Training repeats the loop on small **batches** (a spoonful of 32 dots is one step); one pass through all the data is an **epoch**. The moons network goes from 24% right to 97% in 2,800 steps. Read the practice and test curves together: falling together is healthy; barely moving needs a bigger step; spiky needs a smaller step; practice falling while test rises is memorising. The three cures for memorising are **weight decay** (a small tax on big dials, $L + (\lambda/2)\,\Sigma w^2$), **dropout** (random neurons sit out each practice) and **early stopping**.

Part 2 Practice questions

Choose one answer for each. The answers are on the last page.

1. Which of these is a hyperparameter?

A. a neuron's weights
B. the base price b
C. the step size
D. a neuron's bias

2. In the four-step loop, which step changes the dials?

A. guess
B. check
C. blame
D. nudge

3. A neuron has inputs (2, 1), weights (0.5, −1) and bias 0.5. What is its score z?

A. 0.5
B. 1.5
C. −0.5
D. 2.5

4. What is ReLU(−2)?

A. −2
B. 0
C. 2
D. 0.12

5. Why can one neuron never learn XOR?

A. it has too few weights
B. it has no bias
C. its boundary is a straight line, and no straight line separates the XOR dots
D. the S-curve is too flat

6. Without a bend between them, a hundred layers are equivalent to:

A. a hundred straight lines
B. one straight-line layer
C. a decision tree
D. a single ReLU

7. Misses of 1, −2 and 3. What is the average squared miss?

A. about 0.67
B. 2
C. about 4.67
D. 14

8. Old dial 10, slope 2, step size 0.5. What is the new dial?

A. 9
B. 11
C. 10.5
D. 8

9. Nudging a dial by 0.01 raises the loss from 2.00 to 2.05. Its blame, and which way should it turn?

A. 0.05, up
B. 5, up
C. 5, down
D. −5, down

10. Practice score 100%, test score 85%. The best description and a cure:

A. too slow; a bigger step
B. memorising; weight decay or early stopping
C. healthy; keep going
D. too jumpy; a smaller step

Part 3 Answers

1. **C.** The step size is chosen by you before training; the computer never changes it. Weights and biases (and a and b) are dials it learns. *Point 1.*
2. **D.** Nudge is the update: new dial = old dial − step size × slope. Blame only works out how much each dial is to blame. *Point 2.*
3. **A.** 2 × 0.5 + 1 × (−1) + 0.5 = 0.5. *Point 3.*
4. **B.** ReLU turns every negative number into 0. *Point 4.*
5. **C.** One neuron's boundary is always a straight line, and no straight line puts both orange XOR dots on one side. *Point 5.*
6. **B.** Straight-line layers multiply out into one straight-line layer: $W_2(W_1x) = (W_2W_1)x$. *Point 4.*
7. **C.** (1 + 4 + 9) ÷ 3 ≈ 4.67. *Point 7.*
8. **A.** 10 − 0.5 × 2 = 9. *Point 8.*
9. **C.** 0.05 ÷ 0.01 = 5. Positive blame means turning the dial up makes things worse, so turn it down. *Point 9.*
10. **B.** A big gap between practice and test means memorising; weight decay, dropout or early stopping help. *Point 10.*